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Crop performance of Tasar Silkworm Antheraea mylitta Drury (Lepidoptera: Saturniidae) by using different food plants from Kolhapur district of Western Maharashtra

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Abstract : Indian silk industry occupies a predominant position in the world. Its contribution is only 15% of total world production and more than 80% of silk production is contributed by China. In Maharashtra very few sections of the society are well acquainted with tasar and mulberry silk production .Sericulture in Maharashtra has a tremendous potential especially Western Maharashtra has conducive environment and secondly it adjoins to Karnataka the premier Sericulture State of India. Maharashtra is considered as a non-traditional state of sericulture. Sericulture in Maharashtra is not new, it is about three decades old, while in the tribal districts like Bhandara, Chandrapur, Gadchiroli and Gondia tasar cultivation is practiced for last 300 years. In the present study available host plants from Kolhapur region of Western Maharashtra were used for study of *A. mylitta* D. performance .Western Maharashtra is one of the hot spot of world biodiversity heritage, where very rich floral and faunal diversity is observed, about 6 host plants of *A. mylitta* are available with good density however, up to date it is observed that nobody has much focused on promotion of tasar silk and its host plant propagation, maintenance and utilization in Western Maharashtra .

The present study results showed that, indoor early stage rearing of tasar and later stages in open environment would be a better option to get qualitative and quantitative cocoon production of *A. mylitta*. The present study also indicated that, among all the seasons rearing performance was good in autumn over rainy and winter season .Whereas among the food plants, performance for cocoon parameters like single cocoon weight, peduncle length ,shell weight, pupal weight etc., was observed better with *T.tomentosa* as a food plant are *T. arjuna*, *S. robusta* and *Z. jujuba*

Key Words: Sugar Industry, Actinomycetes ,Occupational Health.

INTRODUCTION

In Maharashtra very few sections of the society are well acquainted with tasar and mulberry silk production. (Sathe and Jadhav, 2001). Sericulture in Maharashtra is not new, it is about three decades old, while in the tribal districts like Bhandara, Chandrapur, Gadchiroli and Gondia Tasar cultivation is practiced for last 300 years. (Jadhav et al., 2011) .Tasar culture is a forest based activity practiced by poor and tribals of India. *Antheraea mylita* Drury is a wild sericigenous insect which produce high economic importance silk. India rank second among the tasar producing countries of the world after China. At present country is producing about 23679 MT of silk including mulberry and Vanya silk (wild silk). Mulberry silk share is 18715 MT whereas vanya silk share is 4964 MT i.e.(Tasar -1729, Eri-3116 and Muga -119 MT) respectively. (CSB, 2013)

India has variety of good quality food plants but only 5% plants out of total population of food plants are used for tasar culture. In many States like Jharkhand, Bihar, Assam, Jammu and Kashmir, Madhya Pradesh, Karnataka

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and Kerala and Maharashtra also have good population as well as variety of tasar food plants available but, due to lack of public awareness tasar culture is not practiced or popularized as per the capacity. It is estimated that about 11.16 million hectares of forest trees constitute the primary and secondary food plants of tasar silk worm. The 80% of the tasar flora is covered by *Shorea robusta* and the rest is covered by *Terminalia* species and other host trees only 5% of tasar food plants in India are put to use for tasar rearing (Singh and Mishra, 2003).Food plant variety and its nutritional value highly affect on cocoon yield and seed quantity of tasar silk. *A. mylitta* which feeds on *T. tomentosa* (Asan), *T. arjuna* (Arjun) and *S. robusta* (Sal) and secondarily on *Lagerstromia parviflora, Zizyphus mauritiana*, *Anogcissus latifolia, Syzigium cumini, Lareya arborea* and *Hardwickia binata* (Reddy et al., 2010). *T. arjuna, T. tomentosa, S. robusta*, which are primary food plants species. But some different eco-races inhabiting diverse eco-climatic zones and feeding on different host plants have become host specific and thus monophagous in behavior (Rai et al., 2006).High nutritive leaf enhance the growth of larvae, increases the immune power of larvae as well as improve quality of cocoons produced. Healthy cocoon can be used for good quality seed production and to get more silk. Larval development affects on adult activeness, pupal development and egg fertility. Due to continuous insect plant interaction, the silk worms have adopted different food plants for their survival in the different ecological habitats (Rao, 2003).

A. mylitta is truly wild species, because of this habit, large production of tasar silk is very difficult. However, if this species reared in outdoor conditions with scientific and technological interventions and we can achieve better cocoon production, it is important to know about feeding habit , which food plant is good for the *A. mylitta* species in a present environment conditions. Best host plant may be concluded by observing the larval growth, larval mortality, percentage of egg fertility larval resistance power and cocoon quality in a rearing trial.

In the present study available host plants from Kolhapur region of Western Maharashtra were used for study of *A. mylitta* .Western Maharashtra is one of the hot spot of world biodiversity heritage, where very rich floral and faunal diversity is observed, about 6 host plants of *A. mylitta* are available with good population however, up to date it is observed that nobody has much focused on promotion of tasar silk and its host plant propagation, maintenance and utilization in western Maharashtra . Natural breeding of *A. mylitta* is also observed in these area where a better condition, some human investment, public awareness and scientific technique will be helpful to increase the production of tasar from Western Maharashtra. It will be a good work for sustainable development of this region, biodiversity conservation and to improve the economic conditions of tribal and economical weaker sections.

Kolhapur district spread over 7585 Sq. kms and near about 1672 Sq.kms is covered by forest out of which about 563 sq.km. is a protected forests. Kolhapur district forest is mainly of two types i.e. subtropical evergreen and dry deciduous. Subtropical evergreen forest found trees like Anjan, Jack fruit, Mango, Hirada, Jambul and semi evergreen and moist deciduous forest consists trees like Biba, Ain, Arjun, Kinni, Umber, Kumbhi, Kinjal, Bhava, Asana, Nana etc.

Environmental conditions of Kolhapur district are conducive for growth of flora and fauna and rearing of *A.mylitta*. Temperature in summer rises up to 40°C in April-May (summer season) and comes down to 14°C during winter season. Rainy season is from June to October, rainfall mainly receives from South West monsoon. On an average 700 mm rainfall and an average humidity of 60% is noticed in the district.

MATERIALS AND METHODS

Tasar eggs were procured form Tasar seed production center, Central Silk Board, Bhandara. In indoor conditions 1000 larvae of *A. mylitta* were reared up to 2nd instars equally on four different host plants like *T. arjuna, T. tomentosa, T. cattappa* and *Z. jujuba* in rearing trays (size 3x2 feet LxW and 4inch height) . Required temperature humidity and other conditions maintained accordingly . Later on larvae's were released in natural habitat on different host plants up to spinning stage . Trials were carried out in three different seasons for one year. Care was taken to protect them from pest and predators by applying nylon nets . Afterword's newly formed cocoons were collected from different host plants. Collected cocoons characterized for different characters

by measuring cocoon weight, pupal weight and shell weight ,ERR ,etc., the data was analyzed on the basis of season and food plants.

RESULTS AND DISCUSSION

RESULTS

Results are shown in table number 1 and graphical representation in fig. 1-5.

Results of rearing performance of *A. mylitta* are presented in table No. 1 .The data shows that *A. mylitta* cocoon performance on different food plants and season reflect variations in cocoon traits like single shell weight, absolute silk, pupal weight, peduncle length, cocoon length and cocoon weight significantly different between different host plant and in different seasons.

In the present study it was observed that best quality cocoon prepared by *A. mylitta* on food plant *T. tomentosa* in autumn season. It shows that autumn is best season for cocoon quality as well as cocoon quantity and *T. tomentosa* is the best food plant as compared to other food plants .*T. arjuna* performance as a better food plant to *T. cattapa* and *Z. jujuba*. Out of three seasons, autumn become the best season however life cycle of *A. mylitta* is lengthy in autumn as compared to other two seasons. In rainy season life cycle was shortest as compared to other two seasons. It is also noticed that, maximum cocoon weight was recorded on *T. tomentosa*

DISCUSSION

According to Reddy *et al*., (2010) cocoon yield and seed quality of tropical tasar silkworm, *Antheraea mylitta* depends on the food plant variety and its nutritional status, but farmers prefer to conduct rearing on alternative tasar food plants based on availability and accessibility which helps them managing the time-bound agricultural works for economic advantage. However, the leaf production rate, quantity and season, gestation period of the plant needs consideration in comparison with primary host plants for the commercial feasibility. The silkworm rearing and grainage performance being important for substance of tropical tasar culture as the cocoons generated on lagentosa plant. Though, the overall performance of tasar rearing and grainage behaviour was found better in *T. tomentosa* food plants, the important commercial traits like cocoon weight, shell weight, silk ratio and egg fertility are positive in *L. parviflora* fed cocoons. This indicates the better availability of nutrients required for silk production and egg fertility in *L. parviflora* food plant than *T. tomentosa* food plant, the important commercial traits like cocoon weight, shell weight, silk ratio and egg fertility are positive in *L. parviflora* fed cocoon. This indicates the better availability of nutrients required for production and egg fertility in *L. parviflora* food plant than *T. tomentosa* and hence, the leaf of *L. parviflora* fed cocoons. This indicates the better availability of nutrients required for silk production and egg fertility in *L. parviflora* food plant than *T. tomentosa* and hence, the leaf of *L. parviflora* can be fed during final larval stages. Attain optional silk and seed productivity in *A. mylitta*. In the present study overall cocoon performance was better on *T. termentosa* was other food plants .The highest cocoon weight 11.08 gm recorded in autumn season an *T. termentosa* where as lowest cocoon weight on *Z. jujube* food plant in rainy season. The data pertaining to pupal weight, peduncle and cocoon length, cocoon width was observed better on *T. tomentosa* and in autumn season except shell weight. Highest shell weight of 1.6 gm was recorded in autumn season and with *T. cattappa* rather than *T. tomentosa.* Lowest cocoon weight, shell weight and peduncle length recorded during rainy season on *Z. jujuba* food plant whereas pupal weight, cocoon length and width observed superior on *T. cattappa*.

Dash et al., (1993) studied cocoon crop performance through seasonal rearing of *Antheraea mylitta* Drury larva on three primary food plants Asan (*Terminalia tomentosa* W. & A.), Arjun (*Terminalia arjun* W. & A.) Sal. (*Shorea robusta* Gaertn.) and three secondary food plant, Ber (*Ziziphus jujuba* Gaertn.) Sidha (*Lagetstroemia parviflora* Roxb.) and Dha (*Anoegeissus latifolia* Wall.) indicate better performance in winter crops than those of a rain and autumn season. Sal among primary food plants appeared uneconomical in terms of total cocoon shell (raw silk) production in spite of a superior cocoon formation. Overall performance was superior in asan than all other food plants during all the seasons, a situation not heretofore documented. The gradation of food plant with regard to performance (total raw silk production) was, in decreasing order of productivity Asan, Arjun, Ber, Sal, Sidha.

Singh *et al*., (2011) reported that Tasar silk worm

Antheraea mylitta D. is a polyphagous insect and is reared out door on the food plant like as *Terminalia arjuna, Terminalia tomentosa, Shorea robusta* . Productivity of cocoon in outdoor rearing is poor due to attack of number of pests and predators besides natural vagaries such as, wide fluctuating temperature, heavy rain and stormy wind etc. Attempts were made in the past to increases the cocoon production by adopting various methods of indoor rearing was conducted in wooden tray supported with wooden rearing frames specially designed for young age and late age silkworms. Trials were taken up to determine optimum number of feeds per day for indoor rearing of tasar silkworm. Three feed per day resulted significantly highest effective rate of rearing (30% and 35% during I and II crop respectively) significant increase commercial (cocoon weight and shell weight) was observed under three feedings condition than other feeding treatments. In the present study three feedings are given in chawki rearing by conserving moisture. Rai et al., (2006) studied the influence of host plant (*T. arjuna*) defenses on the evolution of feeding behaviour in the tasar silkworm and mentioned the following results. Both under indoor and outdoor rearing conditions, early instars of *Antheraea mylitta* showed differential preference for eating towards developmentally different leaves of host plant, *T. arjuna*. Rearing of *A. mylitta* in laboratory as a Chawki (up to II moult) resulted comparatively better (survival 71%) uniform larval growth (79%) and larval weight 0.4 to 5 gm in the later stages of rearing (late age) data reveals that young age survival and effective rate of rearing (ERR) was noticed higher when chawki was undertaken in laboratory and later rearing in open environment in contrast to complete outdoor rearing.

It is also observed in the study that, young age rearing in laboratory and later rearing in outdoor environment gave better results and improvement in cocoon traits than complete outdoor rearing. This practice will also be helpful to avoid unfavorable conditions (June-July) rearing first crop, as there is high rainfall and threat from pest and predators in early stages in this region.

Considering overall performance of A. mylitta *T. tomentosa* rank as a primary food plant and as for as season is concerned autumn shows better traits of cocoon in the present study. Hence, it is suggested that *T. tomentosa* should be given preference for rearing of A. mylitta and for large scale plantation under different tasar promotional pockets of projects involving various rearing and plantation programs.

Table: Season, food plant wise rearing performance of *A.mylitta* D.

Food plant	Rearing season	Cocoon weight (gm) mean value	Shell weight (gm) mean value	Pupal weight (gm) mean value	Peduncle length (cm) mean value	Cocoon length (cm)	Cocoon width (mm)
T. tomentosa	Rainy	9.43	1.45	7.98	4.6	4.3	9.3
	Winter	10.12	1.56	8.56	4.7	4.7	9.7
	Autumn	11.08	1.52	9.56	4.9	5.1	10.2
T. arjuna	Rainy	9.11	1.41	7.7	4.3	4.1	9.3
	Winter	10.06	1.50	8.56	4.7	4.8	9.4
	Autumn	10.12	1.50	8.62	4.5	4.8	9.7
T. cattappa	Rainy	9.06	1.39	7.67	4.2	4.0	9.1
	Winter	9.33	1.46	7.87	4.8	4.3	9.2
	Autumn	10.46	1.56	8.9	4.8	4.7	9.7
Z. jujube	Rainy	8.46	1.33	7.13	3.9	4.1	9.2
	Winter	9.12	1.42	7.7	4.3	4.4	9.4
	Autumn	9.86	1.46	8.4	4.5	4.3	9.6

Table - 2:Climatic conditions of Kolhapur district (2012-13)

Season	Temperature (°C)		Humidity (%)		Rainfall (mm)	
	Minimum	Maximum	Minimum	Maximum	Minimum	Maximum
Rainy	25	32	60	70	100	700
Winter	14	33	58	65	10	40
Autumn	33	40	30	60	0	30

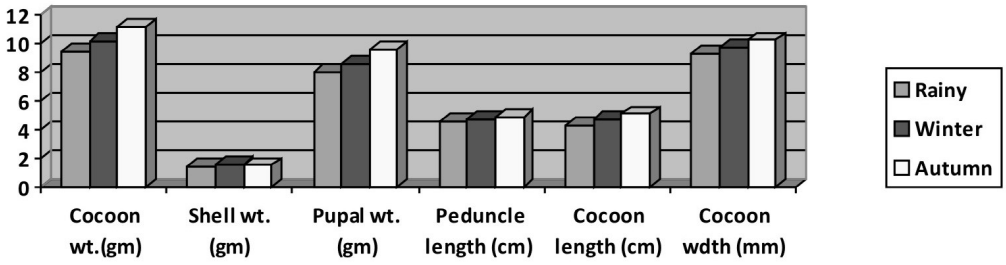


Figure 1. Cocoon parameters from *T.tomentosa* as a food plant

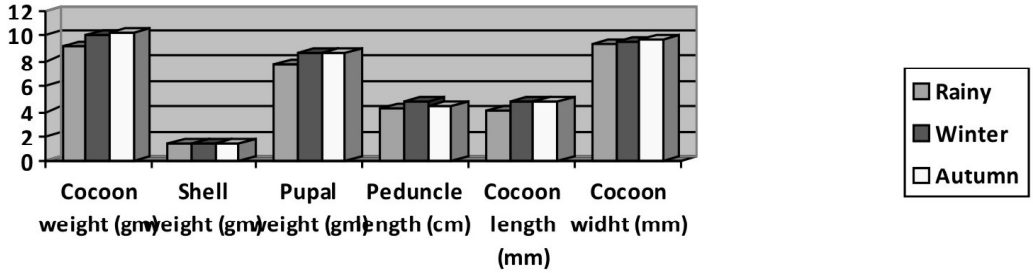


Figure 2. Cocoon parameters from *T. arjuna* as a food plant

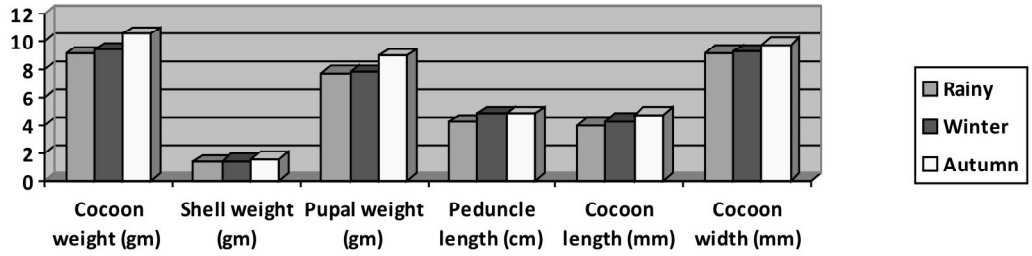


Figure 3. Cocoon parameters from T- *Cattappa* as a food plant

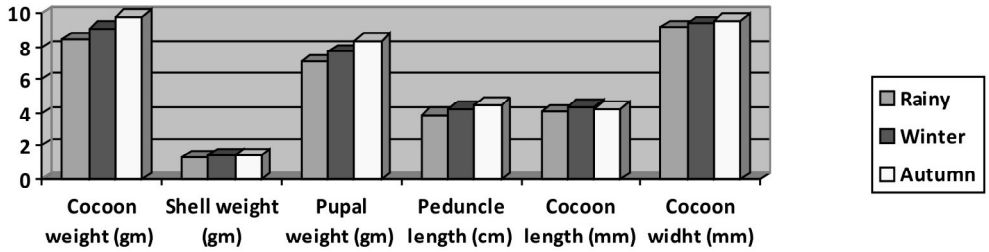


Figure 4. Cocoon parameters from *Z. jujuba* as a food plant

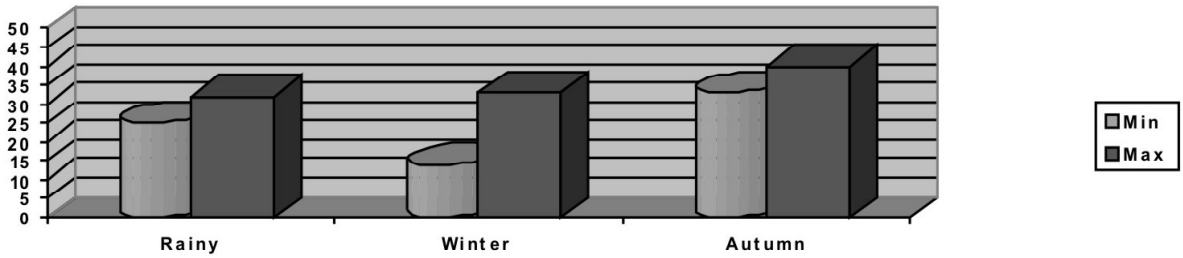


Figure 5. Meteorological data of Kolhapur district

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